

PineRE Current Assessment and Focus

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Version: Tartan 0.01

Status: Active

Last Updated: June 2026

> Documentation Confidence Levels

>

> [CERTAIN] Verified by testing, source code, datasheets, or direct observation.

>

> [LIKELY] Strong supporting evidence exists, but PineRE-specific validation is incomplete.

>

> [UNCERTAIN] Evidence exists, but verification is currently insufficient.

>

> [ASSUMPTION] Planning assumption used until evidence is available.

>

> [SPECULATION] Idea or theory. Not suitable for engineering decisions without validation.

Current View of PineRE

Project Maturity

[CERTAIN]

PineRE has progressed beyond the idea stage.

The project now has:

- Frozen architecture
- Defined roadmap
- Documentation standards
- Decision tracking
- Development workflow

The project is now in an engineering phase rather than an experimentation phase.

Historical Perspective

Alpha

Purpose:

Exploration and proof of concept.

Confidence:

[ASSUMPTION] and [LIKELY]

Alpha demonstrated possibilities but lacks sufficient documentation to be considered authoritative.

Claude

Purpose:

Foundation building.

Confidence:

[CERTAIN] for repository organisation and documented work.

[UNCERTAIN] for hardware behaviour not reproduced.

Claude established the foundations that Tartan builds upon.

Tartan

Purpose:

Engineering and validation.

Confidence:

[CERTAIN]

Tartan should become the first fully reproducible stage of PineRE development.

Current Assessment

| Area | Assessment |

|-----|-----|

| Vision | Excellent |

| Architecture | Frozen |

| Documentation | Good |

| Boot Reliability | Unknown |

| Hardware Support | Unknown |

| Long-Term Potential | Extremely High |

Current Focus

Tartan v0.01

Primary Objective:

Produce a reproducible Alpine login prompt.

Success Criteria:

SD Card

-> BootROM

-> idbloader.img

-> u-boot.itb

-> extlinux.conf

-> Image

-> Device Tree

-> Alpine Linux

-> login:

What Is Not Important Right Now

The following items are intentionally deferred:

- Waydroid

- XFCE

- WiFi

- Bluetooth

- LED Rings

- Theme Systems

- OTA Updates

These features are valuable but should not delay first boot.

Immediate Tasks

1. Verify SD boot path.

2. Verify U-Boot execution.

3. Verify kernel loading.

4. Verify DTB loading.

5. Reach Alpine login prompt.

6. Reproduce boot consistently.

7. Freeze Tartan v0.01.

Engineering Principles

Principle 1

If it cannot be reproduced, it has not been solved.

Principle 2

Documentation is part of development.

Principle 3

Previous versions are reference material, not truth.

Principle 4

Boot first.

Freeze architecture.

Validate hardware.

Polish later.

Looking Ahead

Once Tartan v0.01 is complete:

Tartan v0.02 will focus on:

- Display validation
- Controller validation
- Battery reporting
- USB functionality
- Audio functionality
- DTS freeze

Only after hardware validation should major feature work continue.

Final Assessment

[CERTAIN]

The most important milestone in PineRE is not EmulationStation, Waydroid, or XFCE.

The most important milestone is achieving a reliable and reproducible Alpine login prompt.

That is the point where PineRE becomes a platform instead of an experiment.

Revision History

| Version | Notes |

|-----|-----|

| Tartan 0.01 | Initial assessment and focus document |